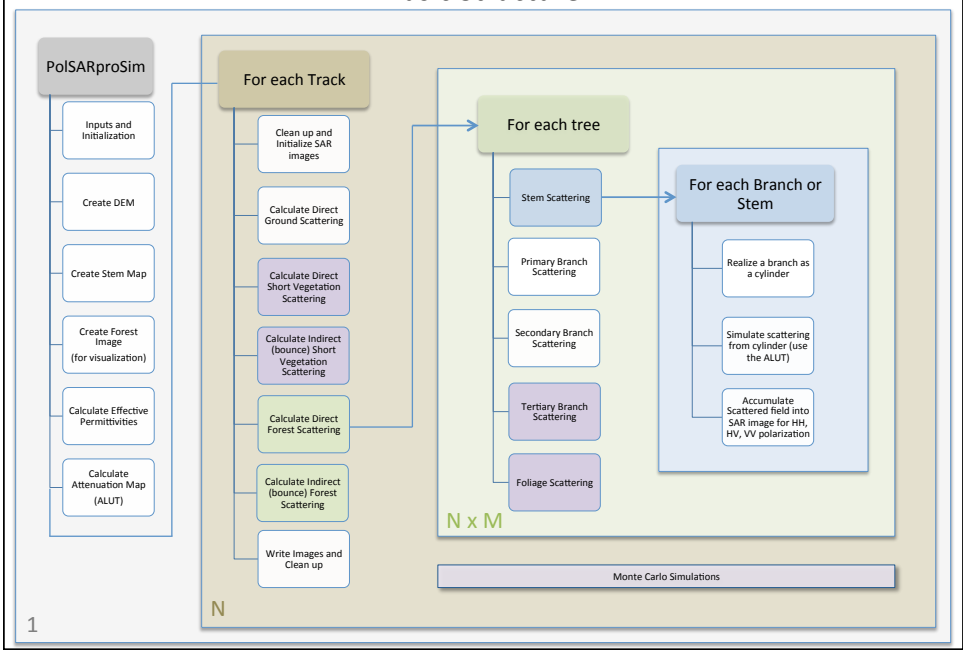
PolSARproSim

Code walkthrough

Basic Structure



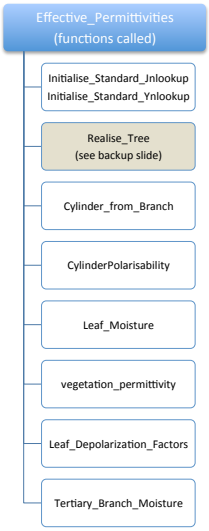
Data Structures

- One Master Record with
  - Various scalar parameters, filenames, file handles etc.
  - Pointers to HH, HV and VV images
  - Pointers to attenuation map, DEM
  - A Record (structure) of Tree parameters such as location, height, dbh etc
  - Record (structure) of Allometric parameters
- A structure for each tree with
  - Scalar parameters (height, dbh etc)
  - Link lists for
    - Crowns (of type crown)
    - Stems (of type branch)
    - Primary branches (of type branch)
    - Dry primary branches (of type branch)
    - Secondary branches (of type branch)
    - Tertiary branches (of type branch)
    - Foliage (of type leaves)

Effective_Permittivities(&Master_Record);

in PolSARproSim_Procedures.c

- Computes the effective wave vectors that are later used to calculate attenuation from a tree crown.
- Loops through each tree and its respective stem, primary, secondary, tertiary branches and leaves and sums the polarizability tensor 'Alpha' for each.
- The effective permittivity tensor of each component is estimated by normalizing the cumulative polarizability tensor by the total crown volume. Effective permittivity of each component is then added up to estimate the effective permittivity of material inside the crown.
- The effective wave vectors are then estimated using effective permittivities.
- Similar process for short vegetation.

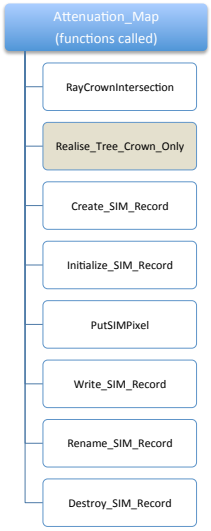


Attenuation_Map (&Master_Record)

in PolSARproSim_Procedures.c

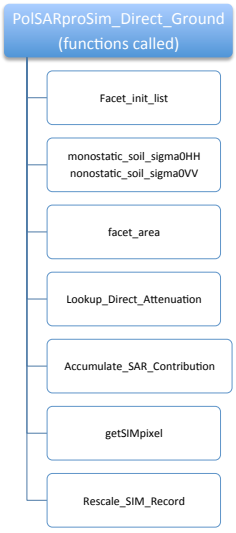
- Populates the Attenuation Map structure in the Master Record.
- This structure contains six lookup tables for wave attenuation
  - Direct H and V
  - Bounce H and V
  - Direct Short Vegetation H and V
- Loops through each voxel, finds the path length of a ray originating at that voxel and traveling towards the radar (for bounce) or towards the ground (for direct) through all intersecting canopies.
- Attenuation is computed using path lengths and wave vectors computed in Effective_Permittivity() as

$$\gamma_h = e^{-|\Im\{K_{oz}\}|\cos\theta\, l_p}$$
$$\gamma_v = e^{-|\Im\{K_{ez}\}|\cos\theta\, l_p}$$



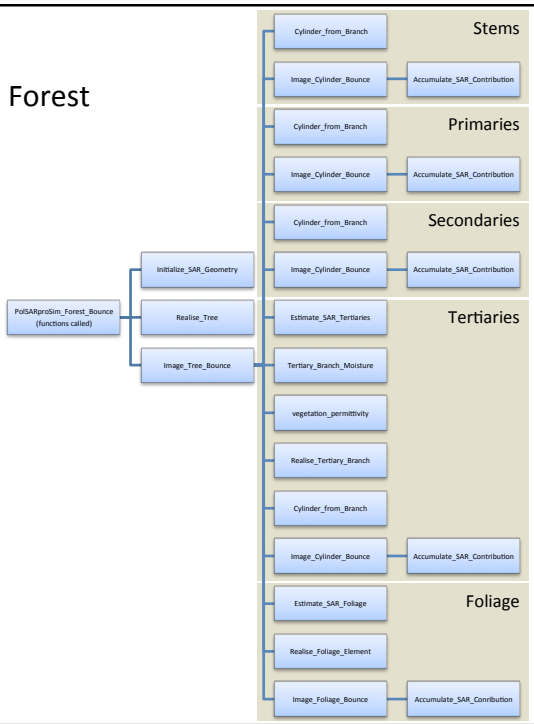
Direct Ground Scattering

- Estimates Radar scattering from ground surface.
- Each resolution element is divided into many triangular facets that can have arbitrary slopes.
- For each triangular facet
  - If the facet is visible, then
    - Calculate the scattered matrix using the SPM model and add noise to its amplitude and phase
    - Scale with facet area, rotate about LOS using Lee et al. and incorporate attenuation using the lookup table
    - Locate the pixel to which this facet belongs and accumulate the scattered field to that particular pixel of the SAR image
    - Normalize and rescale entire image using PSF



Bounce Scattering from Forest

- Estimates the bounce scattering from Forest
- Deterministic/Monte-Carlo approach
- For each tree
  - Realize the tree
  - Image the tree
    - For each stem, Divide the stem into Ns sections, image each section as a finite cylinder, incorporate attenuation, accumulate field
    - For each primary and secondary branch (separately), divide branch into Nb sections, image each section as a finite cylinder, incorporate attenuation, accumulate field
    - For a sample of tertiary branches, divide the cylinders into Nt sections, image each section as a finite cylinder, scale by sample fraction, attenuate using ALUT and accumulate field.
    - For a sample of leaves, image the leaves/needles based on random orientation and location of leaves within the canopy, incorporate attenuation and accumulate field



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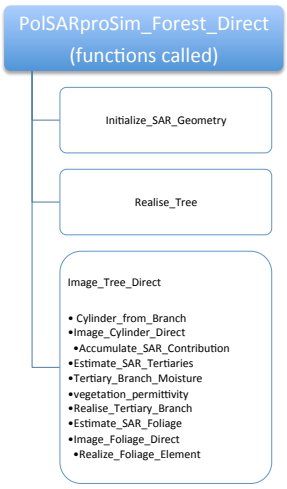
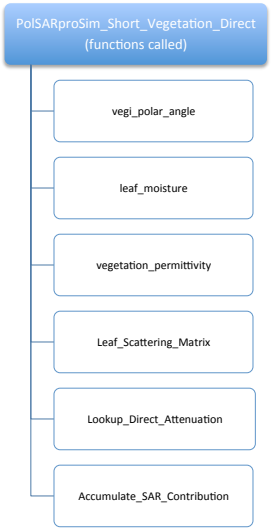


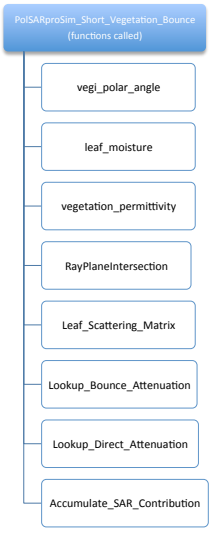
Direct Scattering from Short Vegetation

- Estimates the direct scattering from understory (short vegetation)
- For each pixel in the image
  - For each Monte-Carlo iteration (number based on stem volume fraction)
    - Generate stem of some mean height, and estimate the scattering matrix parameterized by global stem depolarizations and permittivities
    - Scale the scattered field and accumulate in the image
  - For each Monte-Carlo iteration (number based on leaf volume fraction)
    - Generate leaves of some mean extent, and estimate the scattering matrix parameterized by global leaf depolarizations and permittivities
    - Scale the scattered field and accumulate in the image



Indirect Scattering from Short Vegetation

- Estimates the indirect (bounce) scattering from understory (short vegetation)
- Same programming structure as PolSARproSim_Vegetation_Direct, the calculations for scattered field are however different



Direct Ground Scattering, the math

Facet surface normal and unit look vector

$$\hat{n} = n_x\hat{x} + n_y\hat{y} + n_z\hat{z} \qquad \hat{l} = -\sin\theta\hat{y} + \cos\theta\hat{z}$$

$$\cos\theta_i = \hat{n}\cdot\hat{l}$$

$$\beta_f = \tan^{-1}\left(\frac{-n_x}{n_y\cos\theta_i + n_z\sin\theta_i}\right) \qquad \beta' = \beta_X + \beta_f$$

$$S_{hh_l} = \sqrt{fA\sigma^0_{hh}e^{j\phi}} \quad S_{vv_l} = \sqrt{fA\sigma^0_{vv}e^{j\phi}}$$

$$f \sim \mathcal{E}(0,1) \qquad x \sim \mathcal{U}(0,1) \qquad \phi = 2\pi(x-0.5) \qquad \beta_X = 2\Delta\beta(x-0.5)$$

$$\sigma^0_{hh} = k^4\sigma_h^2 c_{len}^2\mathcal{R}\left\{\left|\frac{1-\epsilon}{\sqrt{\epsilon-\sin^2\theta}+\cos\theta}\right|^2 4\cos^4\theta e^{-k^2\sin^2\theta c_{len}^2}\right\}$$

$$\sigma^0_{vv} = k^4\sigma_h^2 c_{len}^2\mathcal{R}\left\{\left|\frac{(\epsilon-1)\left(\epsilon+\epsilon\sin^2\theta-\sin^\theta\right)}{\left(\sqrt{\epsilon-\sin^2\theta}+\epsilon\cos\theta\right)^2}\right|^2 4\cos^4\theta e^{-k^2\sin^2\theta c_{len}^2}\right\}$$

$$\begin{pmatrix} S_{hh} & S_{hv} \\ S_{vh} & S_{vv}\end{pmatrix} = \begin{pmatrix}\cos\beta' & -\sin\beta' \\ \sin\beta' & \cos\beta'\end{pmatrix}\begin{pmatrix} S_{hh_l} & 0 \\ 0 & S_{vv_l}\end{pmatrix}\begin{pmatrix}\cos\beta' & \sin\beta' \\ -\sin\beta' & \cos\beta'\end{pmatrix}$$

Vegetation Permittivity from Moisture, [Ulaby, El-Reyes, 1987]

$$\epsilon_v = \epsilon_r + \epsilon_f v_f + \epsilon_b v_b$$

$$\epsilon_r = 1.7 - 0.74M_g + 6.16M_g^2$$

$$v_b = \frac{4.64M_g^2}{1+7.36M_g^2}$$

$$v_f = M_g(0.55M_g - 0.076)$$

$$\epsilon_f = \left[\epsilon_{f\infty} + \frac{\epsilon_{fs}-\epsilon_{f\infty}}{1+(f/f_{f0})^2}\right] - j\left[\frac{f/f_{f0}(\epsilon_{fs}-\epsilon_{f\infty})}{1+(f/f_{f0})^2} + \frac{g\sigma}{2\pi\epsilon_0 f}\right]$$

$$\epsilon_b = \epsilon_{b\infty} + \frac{\epsilon_{bs}-\epsilon_{b\infty}}{1+\sqrt{jf/f_{b0}}}$$

$$g = 0.0012f^2 + 0.0019f$$

$$----------$$

$$\epsilon_{fs} = 80.88,\ \epsilon_{f\infty} = 4.9,\ \epsilon_{bs} = 57.9,\ \epsilon_{b\infty} = 2.9$$

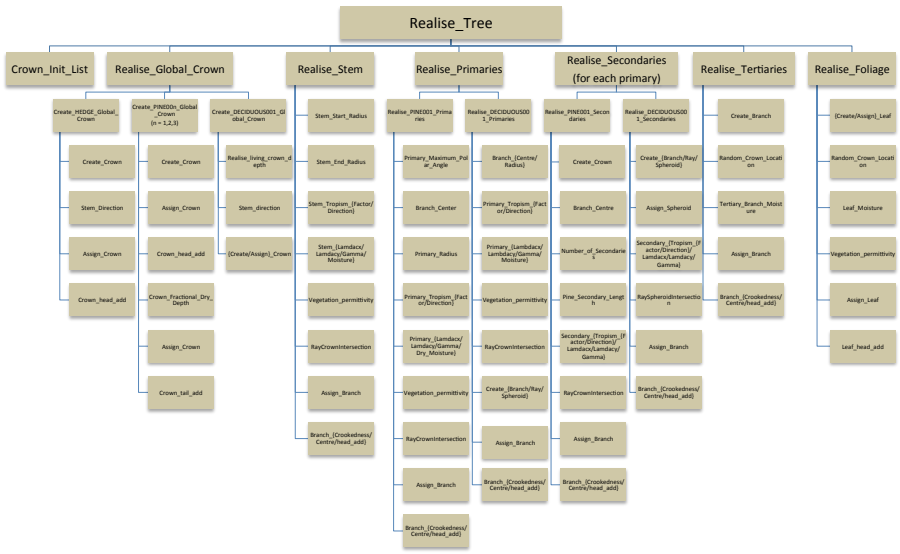
$$f_{f0} = 17.157,\ f_{b0} = 0.18$$

$$\sigma = 1.27,\ \epsilon_0 = 8.854\times10^{-12}$$

$$M_g = \text{Gravimetric Moisture}$$

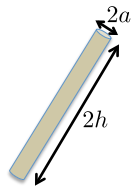
$$f = \text{Radar Frequency (GHz)}$$

Realise_Tree : populates the tree structure



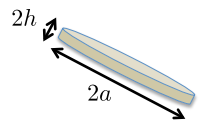
Scattering from a cylinder

- Each branch is simulated as either an infinite or a finite cylinder based on a flag (by default set to simulate an infinite cylinder)
- In case the cylinder is finite
  - Karam, Fung and Antar, "*Electromagnetic Wave Scattering from Some Vegetation Samples*", TGRS 1988
  - Uses generalized Rayleigh-Gans (GRG) Approximation:



$$2ka\sqrt{\epsilon_r} << 1 \quad \text{with} \quad a << h$$

$$\implies a << \frac{\lambda}{4\pi\sqrt{\epsilon_r}}$$



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